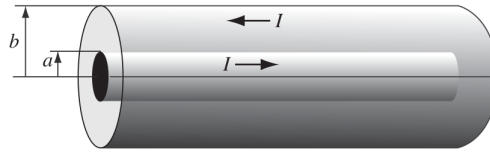


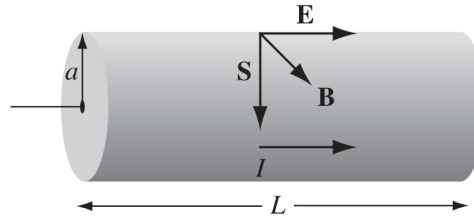
Relativity and Electromagnetism Exercise 4: 2026 Mar 05

Problem 1. What is the electrostatic field outside and inside a uniformly charged spherical shell of total charge q and radius R ? Calculate the energy stored in this electrostatic field. What happens to the energy as $R \rightarrow 0$?

Problem 2. A long coaxial cable carries current I (the current flows down the surface of the inner cylinder, radius a , and back along the outer cylinder, radius b). Find the magnetic energy stored in a section of length l .



Problem 3. Consider a resistive wire in the shape of a cylinder of length L and radius a . A potential difference V is maintained across the wire, and a current I flows through it.



The Poynting vector is defined as $\mathbf{S} \equiv \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$. Calculate the magnitude of Poynting vector at the surface of the wire. In which direction does it point? Obtain the flux of the Poynting vector $\int \mathbf{S} \cdot d\mathbf{a}$ over the surface of the cylinder. What does the result signify?

Problem 4. A monochromatic plane electromagnetic wave travelling in the z direction is described by the electric and magnetic fields

$$\begin{aligned} \mathbf{E}(z, t) &= E_0 \cos(kz - \omega t + \delta) \hat{\mathbf{x}}, \\ \mathbf{B}(z, t) &= \frac{1}{c} E_0 \cos(kz - \omega t + \delta) \hat{\mathbf{y}}. \end{aligned}$$

Calculate

- a. The time average of the electromagnetic energy density in the fields $\langle u_{\text{EM}} \rangle$.
- b. The time average of the Poynting vector $\langle \mathbf{S} \rangle$.